Certified Coverage of Exponential Search Spaces

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Idea. Consider a search problem whose naive state space has size 2^n. Some states may admit a compact structural representation; others may be proposed for elimination by a learned model and certified exactly; still others may be represented by exact summaries of paired groups. Only the residual must be searched explicitly. The central quantity is the *growth of what remains uncovered*, together with the cost of obtaining coverage.

A coverage model

Let $n \in \mathbb{N}$ be the problem size and $\mathcal{S}_n$ its set of candidate states, with

$$|\mathcal{S}_n| = 2^n.$$

A state is *covered* when an exact representation or a sound exclusion removes the need to search it individually. We use three mechanisms on disjoint parts of the space; all state counts below count distinct states.

First, a structural procedure identifies a set $K_n \subseteq \mathcal{S}_n$ of size

$$k(n) = |K_n|$$

that admits a compact representation. Assume $1 \le k(n) \le 2^n$, so the logarithm below is defined. In the simplest logarithmic model, those $k(n)$ states are replaced by a description of cost

$$C_{\text{comp}}(n) = \log_2 k(n).$$

This is deliberately an abstract assumption: it describes the cost once such a representation is available, not a universal property of arbitrary subsets.

Second, a learned classifier proposes regions outside K_n for removal. Let $\delta(n)$ count the distinct proposed states, counting overlaps only once. The classifier cannot delete states directly: a sound verifier certifies only regions whose omission leaves the search answer unchanged. Let $a(n) \in [0, 1]$ be the fraction of those $\delta(n)$ states certified by the verifier, with $a(n) = 0$ if $\delta(n) = 0$. The safely eliminated count is

$$\delta_c(n) = a(n)\,\delta(n).$$

Prediction error therefore affects efficiency, not correctness: uncertified proposals simply remain in the residual problem.

Third, pair groups of remaining states and replace each pair by an exact summary for the search question. Let $p(n)$ count the distinct states represented by these paired summaries, excluding states already counted in $k(n)$ or $a(n)\delta(n)$. Let $Q_{\text{pair}}(n)$ be the cost of finding pairs, constructing and certifying their summaries, and using the resulting representation.

The explicit residual is

$$r(n) = 2^n - k(n) - a(n)\delta(n) - p(n). \tag{1}$$

Let $P(n)$ collect structural discovery and learned proposal and verification work, excluding the pairing cost $Q_{\text{pair}}(n)$ and the structural description charge $\log_2 k(n)$. All costs use the same units. Normalizing explicit search to one unit per residual state gives the modeled runtime

$$T(n) = P(n) + Q_{\text{pair}}(n) + r(n) + \log_2 k(n). \tag{2}$$

Thus

$$\begin{aligned} T(n) = P(n) + Q_{\text{pair}}(n) + 2^n - k(n) \\ - a(n)\delta(n) - p(n) + \log_2 k(n). \end{aligned} \tag{3}$$

The separation is important. Large covered regions help only if discovering, representing, and certifying them is cheap.

The certified coverage theorem

Define the total *certified coverage*

$$C(n) = k(n) + a(n)\delta(n) + p(n),$$

so that $r(n) = 2^n - C(n)$. Here and below, $f(n) = \text{poly}(n)$ means $f(n) = O(n^d)$ for some fixed $d \geq 0$.

Theorem 1 (Certified coverage criterion). Assume $P(n) + Q_{\text{pair}}(n) = \text{poly}(n)$, $1 \leq k(n) \leq 2^n$, and the residual is searched in time proportional to $r(n)$. Then

$$T(n) = \text{poly}(n) \quad \Longleftrightarrow \quad 2^n - C(n) = \text{poly}(n).$$

Equivalently, polynomial modeled runtime occurs exactly when

$$C(n) = 2^n - \text{poly}(n).$$

Proof. From (2),

$$T(n) = P(n) + Q_{\text{pair}}(n) + \big(2^n - C(n)\big) + \log_2 k(n).$$

Because $k(n) \leq 2^n$,

$$0 \leq \log_2 k(n) \leq n.$$

The overhead and logarithmic terms are polynomially bounded. Hence $T(n)$ is polynomial exactly when the residual $2^n - C(n)$ is polynomially bounded. □

The theorem gives a sharp asymptotic warning: eliminating a very large *percentage* of an exponential space need not change its asymptotic order.

Corollary 1 (Fixed uncovered fraction). If there exists a constant $\varepsilon > 0$ such that, for infinitely many n,

$$C(n) \leq (1 - \varepsilon)2^n,$$

then, for those n,

$$T(n) \geq r(n) \geq \varepsilon 2^n.$$

Thus 90%, 99%, or even 99.999% coverage is asymptotically insufficient if the uncovered fraction stays bounded away from zero. Polynomial behavior requires

$$\frac{C(n)}{2^n} = 1 - \frac{\mathrm{poly}(n)}{2^n},$$

so the uncovered fraction must vanish exponentially fast, up to polynomial factors.

Differential form

For intuition, choose differentiable extensions of k, δ, a, p, P, and Q_{pair} from integer to real n, with $k > 0$, and define r, T there by (1)–(2). Primes denote derivatives with respect to n; the equations below describe this chosen interpolation. Differentiating (1) gives

$$\frac{dr}{dn} = (\ln 2)2^n - k'(n) - a(n)\delta'(n) - a'(n)\delta(n) - p'(n). \tag{4}$$

The terms are raw state creation, structural coverage, new certified proposals, improving certification rate, and paired coverage, respectively.

In the next two displays, unprimed k, a, δ abbreviate their values at n. Differentiating the full cost gives

$$\begin{aligned}\frac{dT}{dn} = (\ln 2)2^n - \left(1 - \frac{1}{k \ln 2}\right) k' - a\delta' - a'\delta - p' \\ + P'(n) + Q'_{\mathrm{pair}}(n).\end{aligned} \tag{5}$$

When the correction $k'/(k \ln 2)$ is small relative to the terms of interest, this gives the approximate balance law

$$\begin{aligned}T'(n) \approx (\ln 2)2^n - k' - a\delta' - a'\delta - p' \\ + P'(n) + Q'_{\mathrm{pair}}(n).\end{aligned} \tag{6}$$

Equivalently, since $r(n) = 2^n - C(n)$,

$$r'(n) = (\ln 2)2^n - C'(n).$$

Integrating from a reference scale n_0,

$$r(n) = r(n_0) + \int_{n_0}^{n} \left[(\ln 2)2^s - C'(s)\right] ds. \tag{7}$$

This form makes the central picture transparent: at integer n, polynomial behavior requires the cumulative gap between exponential state creation and certified coverage growth to remain polynomial in n.

Why certification matters

Suppose a naive learned-pruning model deletes $\delta(n)$ states and $e(n) \in [0, 1]$ is the fraction deleted incorrectly. Exact recovery would require revisiting $e(n)\delta(n)$ states. When $\delta(n) = \Theta(2^n)$, any fixed positive error fraction leaves exponential recovery work. Keeping that work polynomial requires, for some fixed $d \geq 0$,

$$e(n)\delta(n) = O(n^d), \qquad \text{hence} \qquad e(n) = O\left(\frac{n^d}{2^n}\right).$$

Thus ordinary constant-error prediction is not enough for exact bulk deletion of an exponential region.

Certification avoids that failure mode. The learned model becomes a *proposal mechanism*: it suggests large regions that may be irrelevant, while an exact verifier alone has authority to delete them. A poor classifier can waste computation, but it cannot make the final algorithm incorrect. In this formulation, learning changes δ and the acceptance function a; correctness is carried by verification.

The cost of pairing

For disjoint paired groups $B_0, B_1 \subseteq \mathcal{S}_n$, let $s(B)$ be a summary of group B, and let Merge combine two summaries. Exact pairing requires

$$s(B_0 \cup B_1) = \mathrm{Merge}\big(s(B_0), s(B_1)\big),$$

Here *exact* means that replacing a paired group by its summary leaves the final answer unchanged when combined with all other covered groups and the residual. A complete explicit pairing tree over 2^n states has $2^n - 1$ merges, even though only one group remains. Thus pairing helps asymptotically only when a compact rule or reuse of summaries keeps $Q_{\mathrm{pair}}(n)$ polynomial. Counting covered states without this cost would hide the original exponential work.

Scope and the remaining difficulty

The coverage criterion is a theorem about the model; it does not assert that every concrete search problem admits such coverage. The hard question is whether, for a given problem, one can simultaneously achieve

$$P(n) + Q_{\mathrm{pair}}(n) = \mathrm{poly}(n) \qquad \text{and} \qquad 2^n - C(n) = \mathrm{poly}(n).$$

Cardinality alone cannot answer this: exponentially many states can sometimes be described by a short rule. The substantive lower- or upper-bound problem is therefore about the *cost of discovering and certifying structure*, not merely the number of states represented by that structure.

General representation cost

The logarithmic term is the motivating special case, not an essential part of the coverage theorem. Let $\rho(n, k)$ be the cost to store and use a structural representation covering k states at problem size n. Then write

$$T(n) = P(n) + Q_{\mathrm{pair}}(n) + r(n) + \rho(n, k(n)).$$

Whenever $P(n) + Q_{\mathrm{pair}}(n)$ and $\rho(n, k(n))$ are polynomially bounded, the same proof gives

$$T(n) = \mathrm{poly}(n) \quad \Longleftrightarrow \quad r(n) = 2^n - C(n) = \mathrm{poly}(n).$$

Thus the essential object is the uncovered residual; $\log_2 k$ is simply the strongest compression regime considered here. This also separates two questions that should not be conflated: how many states a representation covers, and how costly that representation is to discover and manipulate.

Summary. Exponential search becomes a competition between state-space growth and certified coverage. Structural compression and exact pairing represent states collectively; learning proposes further regions; certification makes eliminations exact; and the residual determines explicit work. The governing condition is simple:

$$\text{polynomial overhead} + \text{polynomial residual} \implies \text{polynomial modeled runtime.}$$

The differential form records the same statement dynamically: coverage must track exponential state creation closely enough that their accumulated gap is only polynomial.